

10. TRANSFORM TECHNIQUES AND SYSTEM REPRESENTATION

Objective

This experiment is divided into four simulations:

1. Pole Zero Plot
2. Z-Transform and Inverse Z-Transform
3. Z-Transform of the Given Signal
4. Laplace Transform of Continuous-Time Signals

Experiment 10.1

Pole Zero Plot

Program

% Test Case: Manual Pole-Zero Plot

```
clc;
```

```
clear;
```

```
close all;
```

```
% Define the coefficients of the transfer function
```

```
b = [1 -0.5];      % Numerator coefficients (zeros)
```

```
a = [1 -1.2 0.32]; % Denominator coefficients (poles)
```

```
% Calculate poles and zeros
```

```
z = roots(b);
```

```
p = roots(a);
```

```
% Create a new figure
```

```
figure;
```

```
hold on;
```

```
grid on;
```

```
axis equal;
```

```

% Define axis range

re = -2:0.01:2;
im = -2:0.01:2;

% Plot real axis
plot(re, zeros(size(re)), 'k-', 'LineWidth', 1);

% Plot imaginary axis
plot(zeros(size(im)), im, 'k-', 'LineWidth', 1);

% Plot unit circle
theta = linspace(0, 2*pi, 100);
unit_circle_x = cos(theta);
unit_circle_y = sin(theta);
plot(unit_circle_x, unit_circle_y, 'r--', 'LineWidth', 1);

% Plot zeros
plot(real(z), imag(z), 'go', ...
    'MarkerSize', 8, 'LineWidth', 2);

% Plot poles
plot(real(p), imag(p), 'rx', ...
    'MarkerSize', 10, 'LineWidth', 2);

% Customize the plot
xlabel('Real Part');
ylabel('Imaginary Part');
title('Manual Pole-Zero Plot');

legend('Real Axis', 'Imaginary Axis', 'Unit Circle', ...

```

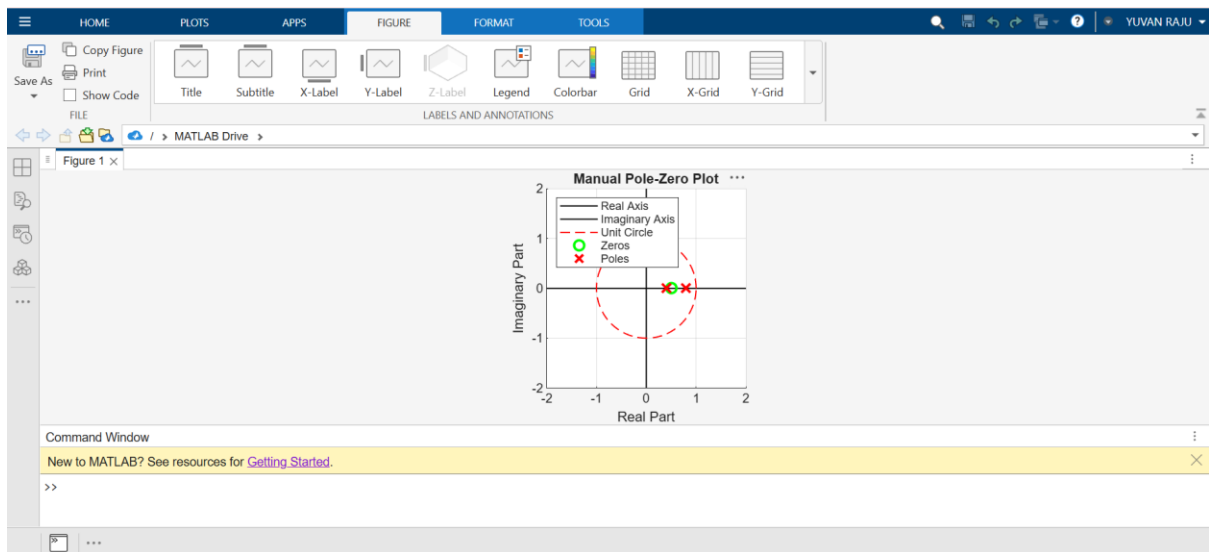
```
'Zeros', 'Poles', 'Location', 'northwest');
```

```
xlim([-2 2]);
```

```
ylim([-2 2]);
```

```
hold off;
```

OUTPUT



Experiment 10.2

Z-Transform and Inverse Z-Transform

Program

```
% =====
```

```
% Numerical Z-Transform
```

```
% =====
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Given sequence
```

```
x = [1 0.5 0.25 0.125];
```

```

% Number of points for DFT
N = 64;

% Time index
n = 0:length(x)-1;

% Define z-points on unit circle
k = 0:N-1;
omega = 2*pi*k/N;
z = exp(1j*omega);

% Compute Z-transform on unit circle
Xz = zeros(1,N);

for m = 1:N
    Xz(m) = sum(x .* (z(m).^(-n)));
end

% =====
% Plot Magnitude and Phase
% =====

figure;

% Magnitude Spectrum
subplot(2,1,1);
plot(omega/(2*pi), abs(Xz), 'LineWidth', 1.5);
xlabel('Normalized Frequency');
ylabel('|X(z)|');
title('Magnitude Spectrum');
grid on;

```

```

% Phase Spectrum
subplot(2,1,2);
plot(omega/(2*pi), angle(Xz), 'LineWidth', 1.5);
xlabel('Normalized Frequency');
ylabel('Phase (rad)');
title('Phase Spectrum');
grid on;

% =====

% Numerical Inverse Z-Transform
% =====

% Recover x[n] using inverse DFT
x_rec = zeros(1,N);

for m = 1:length(n)
    x_rec(m) = (1/N) * sum(Xz .* (z.^n(m)));
end

% Display recovered sequence
disp('Recovered sequence from Inverse Z-Transform:');
disp(real(x_rec(1:length(x))));

OUTPUT

```

